

第二次project记录

Part One : The simplify of system

Consider the complex form of the autonomous systems :

$$A : \dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z}$$

The condition of A becomes a Hamiltonian system is :

$$2A = -\bar{C}$$

We know that :

$$C := C_1 + iC_2$$

Now we do the rotation :

$$z \rightarrow ze^{i\alpha} \stackrel{\text{denote } \sigma := e^{i\alpha}}{=} z\sigma$$

We get A becomes :

$$\tilde{A} : \sigma\dot{z} = i\sigma z + A\sigma^2 z^2 + B\bar{\sigma}^2 \bar{z}^2 + C\sigma\bar{\sigma} z\bar{z}$$

$\Longleftrightarrow$

$$\tilde{A} : \dot{z} = iz + A\sigma z^2 + B\bar{\sigma}^3 \bar{z}^2 + \bar{\sigma}Cz\bar{z}$$

Now the condition for $\tilde{A}$ becomes a Hamiltonian system is :

$$2(\sigma A) = -(\bar{\sigma}C)$$

$\Longleftrightarrow$

$$2A = -\bar{C}$$

This tell us a fact that the condition is preseve under the rotation operation. That is :

We can always choose an σ s.t $C \in \mathbb{R}$

That is :

Without losing generality , We can let $Im(C) = 0$

Therefore , Let $Im(C) = 0$ Then : A becomes in $\mathbb{R}$.

So now : The system $\tilde{A}$ is only depended on A_1, B_1, B_2 We denote :

$$\lambda := (A_1, B_1, B_2)$$

Part Two : The Singular λ

Definition 1 : (*singular*) $\lambda = (A_1, B_1, B_2)$

Suppose that we have a Hamiltonian system A with $H(x, y, \lambda)$ That is :

$$A : \begin{cases} \dot{x} = H_y(x, y, \lambda) \\ \dot{y} = -H_x(x, y, \lambda) \end{cases}$$

Then we say that λ is *Singular* if $H(x, y, \lambda)$ satisfies one of the following conditions :

CASE I : $H(x, y, \lambda)$ has a degenerate critical point as a function of (x, y)

That is : There $\exists (x_0, y_0)$ s.t
$$\begin{cases} H_x(x_0, y_0, \lambda) = 0 \\ H_y(x_0, y_0, \lambda) = 0 \\ \det(\text{Hess}H(x_0, y_0, \lambda)) = 0 \end{cases}$$

CASE II : $H(x, y, \lambda)$ has two different critical points $(x_1, y_1), (x_2, y_2)$ s.t $H(x_1, y_1, \lambda) = H(x_2, y_2, \lambda)$

Definition 2 : $(\Sigma, \Sigma_1, \Sigma_2)$ Given a *singular* $\lambda = (A_1, B_1, B_2)$ We define : i. $\lambda \in \Sigma_1 \iff H(x, y, \lambda)$ is in **CASE I** ii. $\lambda \in \Sigma_2 \iff H(x, y, \lambda)$ is in **CASE II** iii. $\Sigma := \Sigma_1 \cup \Sigma_2$

We call : i. Σ_1 *caustic* We call : ii. Σ_2 *Maxwell stratum*

We call : iii. Σ the bifurcation diagram of the family of $H(x, y, \lambda)$

Example : Consider $H(x) = x^4 + \lambda_1 x^2 + \lambda_2 x$ How to draw Σ ? I.(About Σ_1) :

$$\text{let } \begin{cases} H_x = 4x^3 + 2\lambda_1 x + \lambda_2 = 0 \\ H_{xx} = 12x^2 + 2\lambda_1 = 0 \end{cases}$$

Solve this
 $\Rightarrow$

$$\begin{cases} \lambda_1 = -6x^2 \\ \lambda_2 = 8x^3 \end{cases}$$

This is a parameter equation about x Then we can draw Σ_1 in λ_1 axis and λ_2 axis ! II.(About Σ_2) : Suppose x_a and x_b

$$\text{let } \begin{cases} 4x_a^3 + 2\lambda_1 x_a + \lambda_2 = 0 \\ 4x_b^3 + 2\lambda_1 x_b + \lambda_2 = 0 \\ x_a^4 + \lambda_1 x_a^2 + \lambda_2 x_a = x_b^4 + \lambda_1 x_b^2 + \lambda_2 x_b \end{cases}$$

Solve it
 $\Rightarrow$

$$\begin{cases} \lambda_1 < 0 \\ \lambda_2 = 0 \end{cases}$$

Then we can draw Σ_2 in λ_1 axis and λ_2 axis ! Therefore , the graph of $\Sigma = \Sigma_1 \cup \Sigma_2$ is done !

Remark : (small trick) Let $C_1 = 0$ and let $C_1 = 1$

Part Three : Problem

Find and draw the bifurcation diagram $\Sigma = \Sigma_1 \cup \Sigma_2$